From bond activation to interpretable clot mechanics

(a) Observed activation

C5: 0.4497% strain; 2 / 5,392 bonds
C7: 0.3138% strain; 2 / 46,059 bonds

Sampling changes the apparent fraction.
Mesh objectivity is not established.

↓

(b) Numerical evidence

C5: 13 audited stable positive states
C7: 20 stationary positive states; stability unassessed

Half-reaction / continuum: 1.376 (C5),
1.442 (C7); loading issue unresolved.

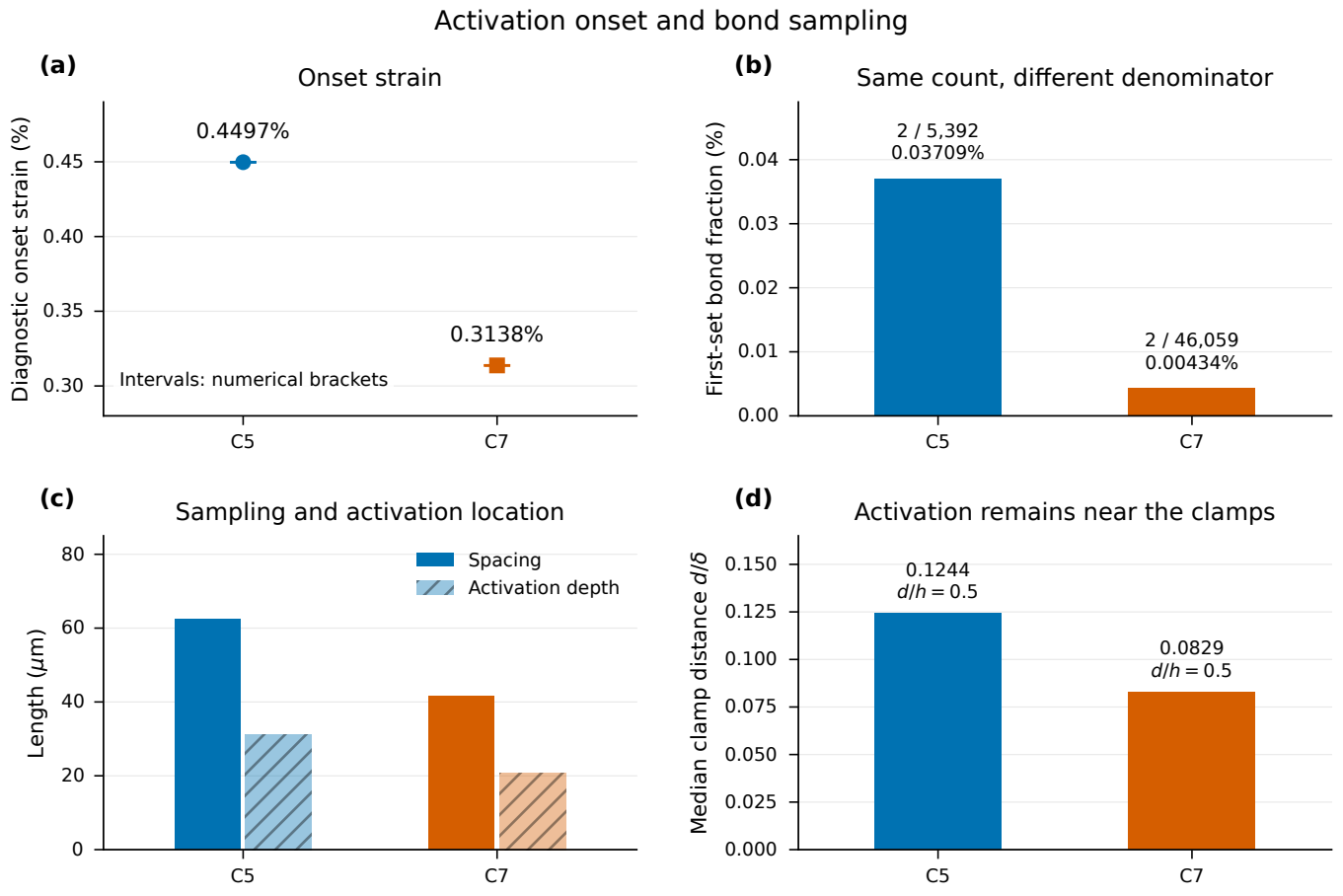
↓

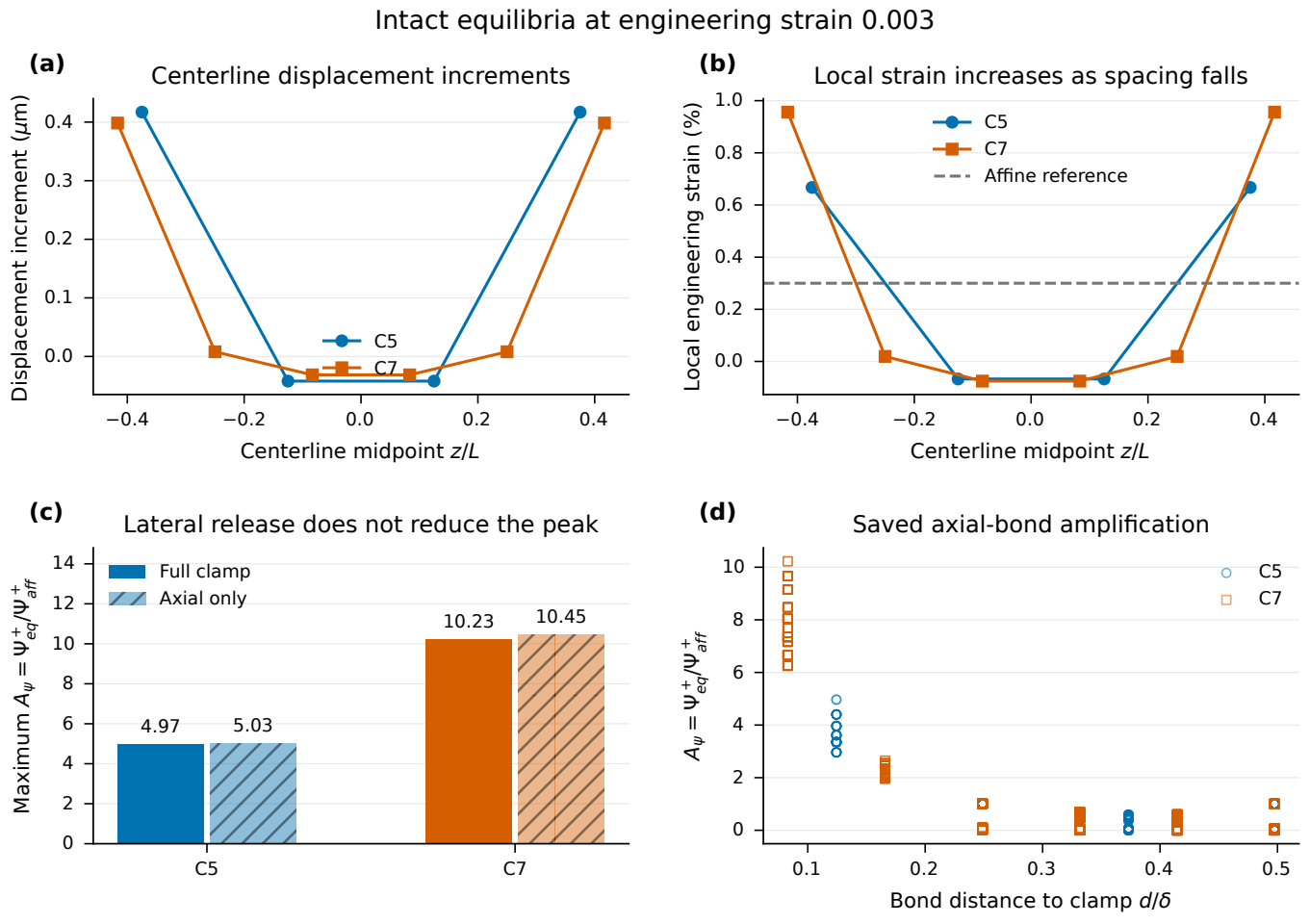
(c) Clot-model implications

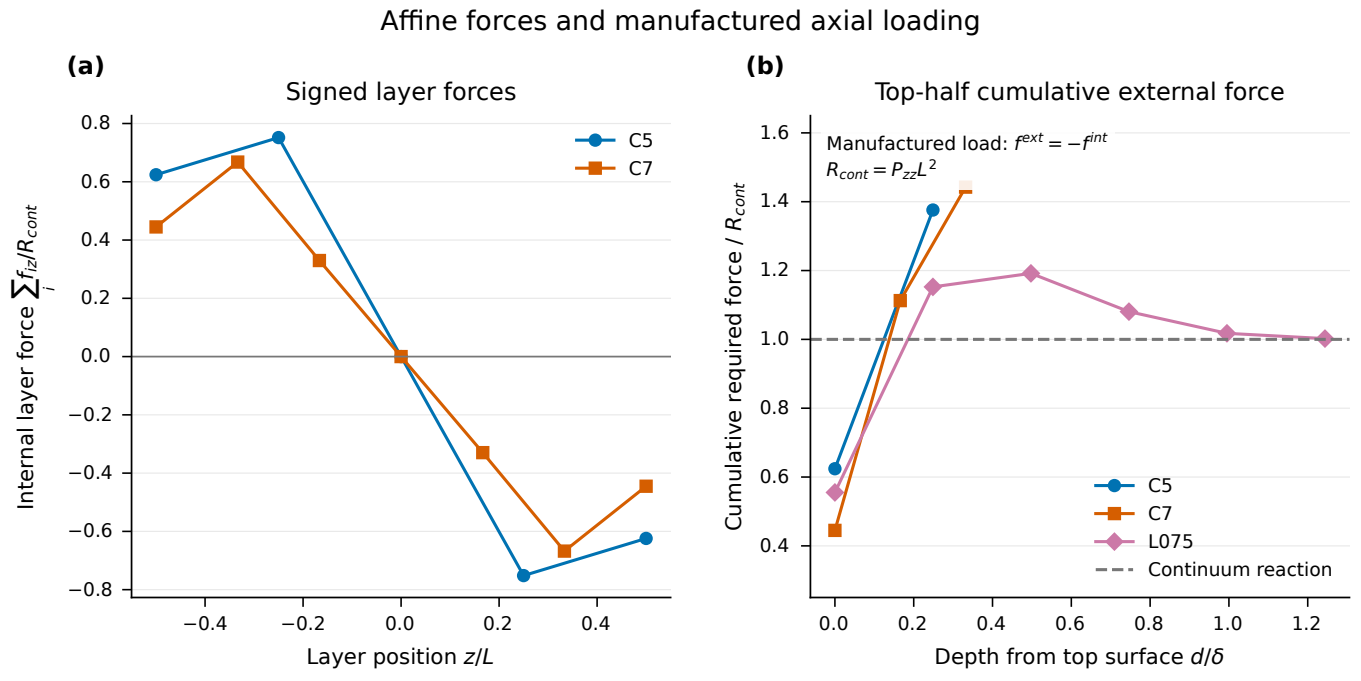
Validate finite-domain loading and force support.
Define a physically interpretable failure observable.

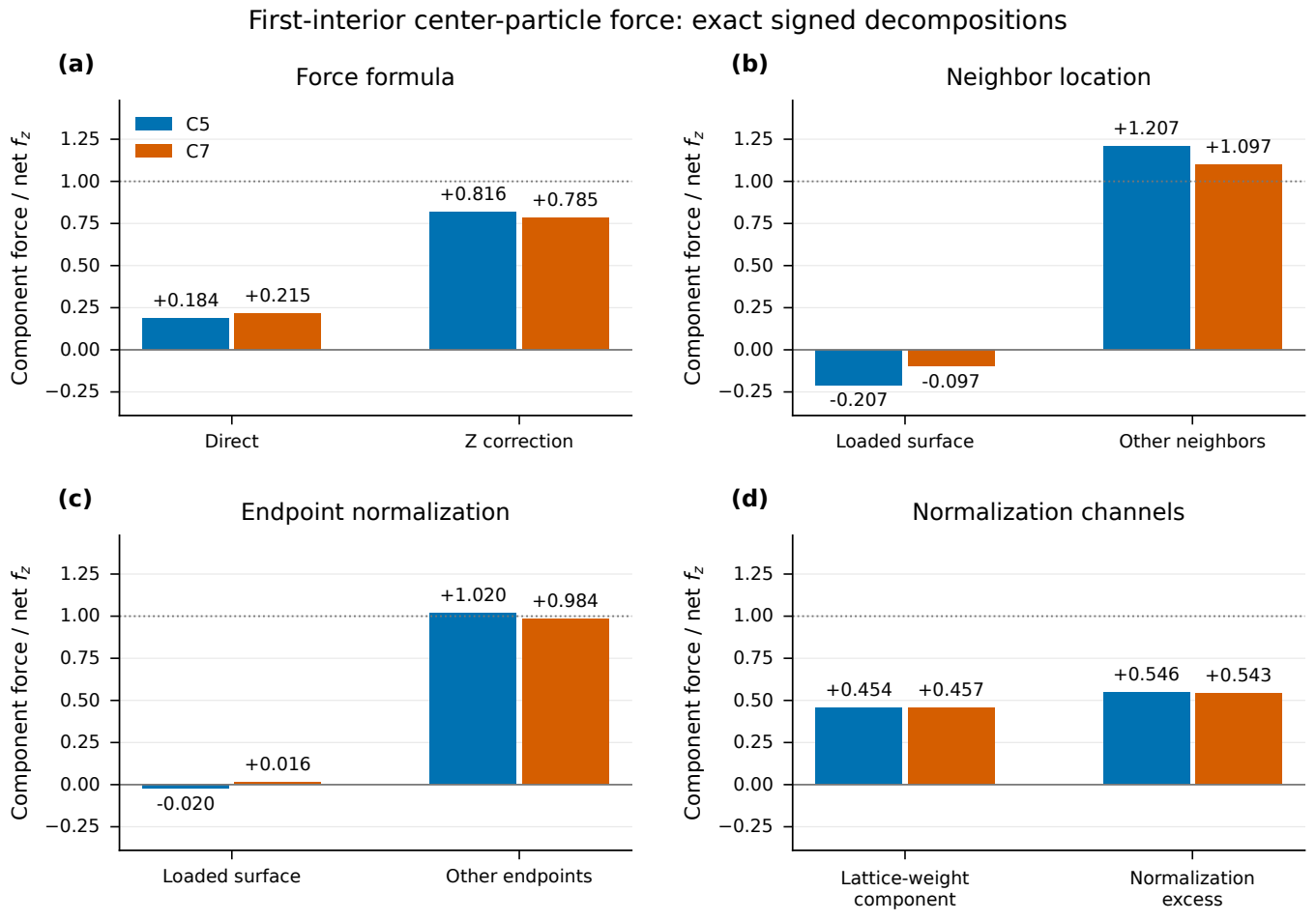
Then test clot-specific calibration and coupling.
Current coupon evidence is not a rupture prediction.

Research argument only; arrows do not establish causality or a biological mechanism.

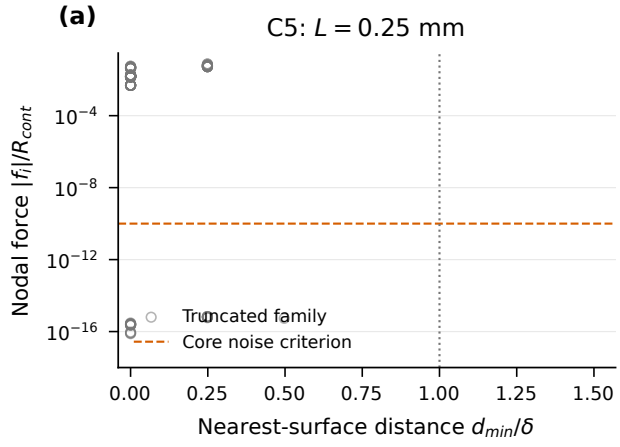

Activation onset and bond sampling
(a)
Onset strain
Diagnostic onset strain (%)
0.4497%
0.3138%
Intervals: numerical brackets
C5
C7
(b)
Same count, different denominator
First-set bond fraction (%)
2 / 5,392
0.03709%
2 / 46,059
0.00434%
C5
C7
(c)
Sampling and activation location
Length (μm)
Spacing
Activation depth
C5
C7
(d)
Activation remains near the clamps
Median clamp distance d/δ
0.1244
d/h = 0.5
0.0829
d/h = 0.5
C5
C7

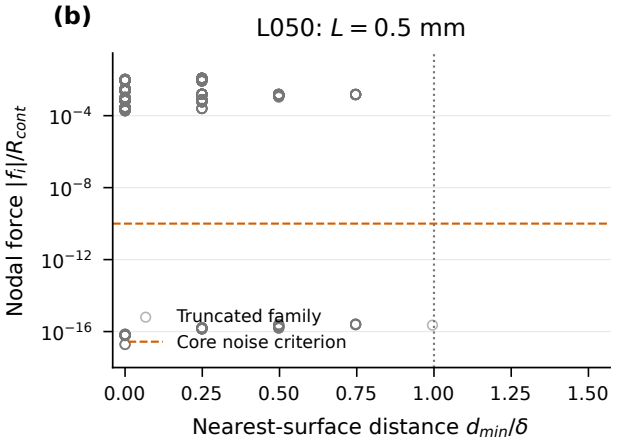

Intact equilibria at engineering strain 0.003
(a)
Centerline displacement increments
Displacement increment (μm)
0.4
0.3
0.2
0.1
0.0
C5
C7
−0.4
−0.2
0.0
0.2
0.4
Centerline midpoint z/L
(b)
Local strain increases as spacing falls
Local engineering strain (%)
1.0
0.8
0.6
0.4
0.2
0.0
C5
C7
Affine reference
−0.4
−0.2
0.0
0.2
0.4
Centerline midpoint z/L
(c)
Lateral release does not reduce the peak
Maximum $A_\psi = \Psi^+_{eq}/\Psi^+_{aff}$
14
12
10
8
6
4
2
0
Full clamp
Axial only
4.97
5.03
10.23
10.45
C5
C7
(d)
Saved axial-bond amplification
$A_\psi = \Psi^+_{eq}/\Psi^+_{aff}$
10
8
6
4
2
0
C5
C7
0.1
0.2
0.3
0.4
0.5
Bond distance to clamp d/δ

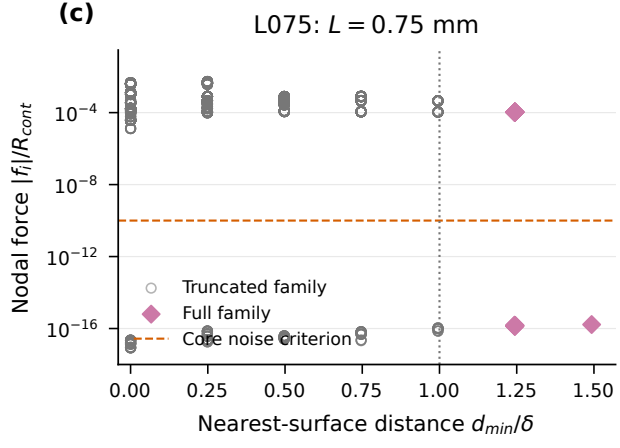

Affine forces and manufactured axial loading
(a)
Signed layer forces
C5
C7
Internal layer force $\sum_i f_{iz}/R_{cont}$
Layer position z/L
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.4
−0.2
0.0
0.2
0.4
(b)
Top-half cumulative external force
Manufactured load: $f^{ext} = -f^{int}$
$R_{cont} = P_{zz}L^2$
Cumulative required force / R_{cont}
Depth from top surface d/δ
C5
C7
L075
Continuum reaction
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.0
0.2
0.4
0.6
0.8
1.0
1.2


First-interior center-particle force: exact signed decompositions
(a)
Force formula
C5
C7
Component force / net f_z
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
+0.184
+0.215
+0.816
+0.785
Direct
Z correction
(b)
Neighbor location
Component force / net f_z
+1.207
+1.097
-0.207
-0.097
Loaded surface
Other neighbors
(c)
Endpoint normalization
Component force / net f_z
+1.020
+0.984
-0.020
+0.016
Loaded surface
Other endpoints
(d)
Normalization channels
Component force / net f_z
+0.454
+0.457
+0.546
+0.543
Lattice-weight component
Normalization excess

Geometric family support and affine force residuals



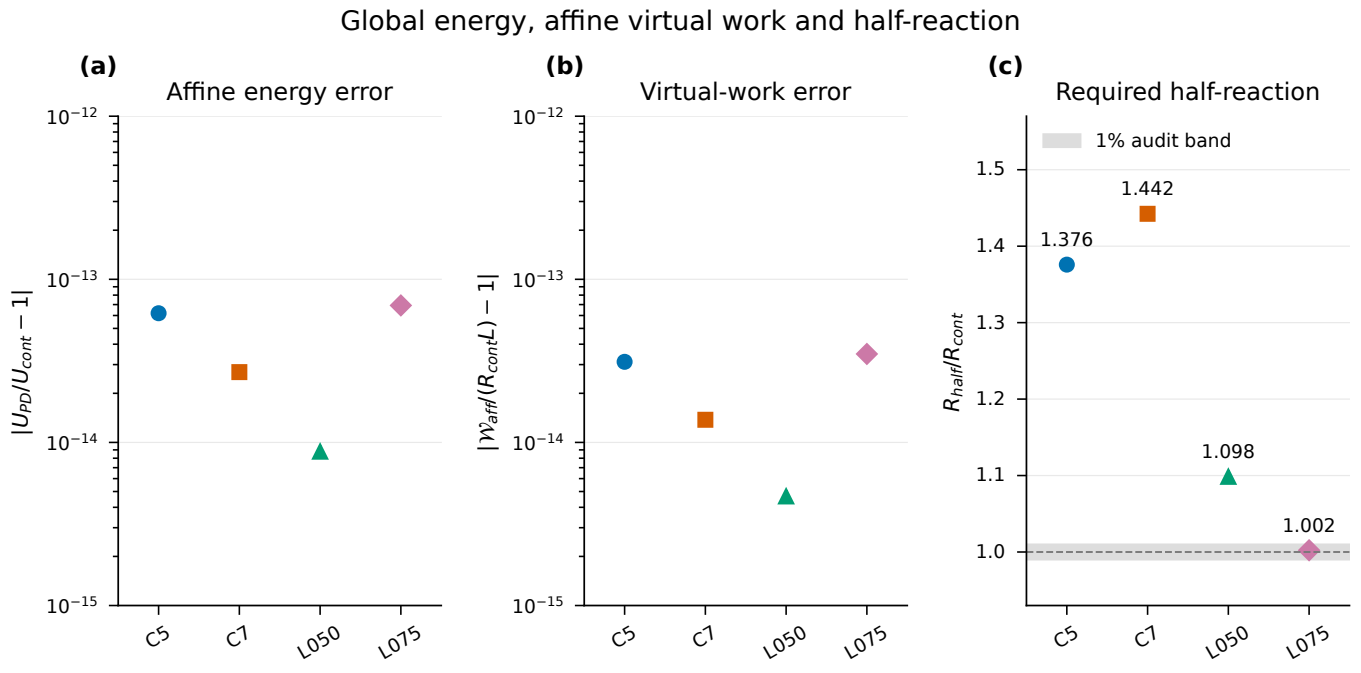


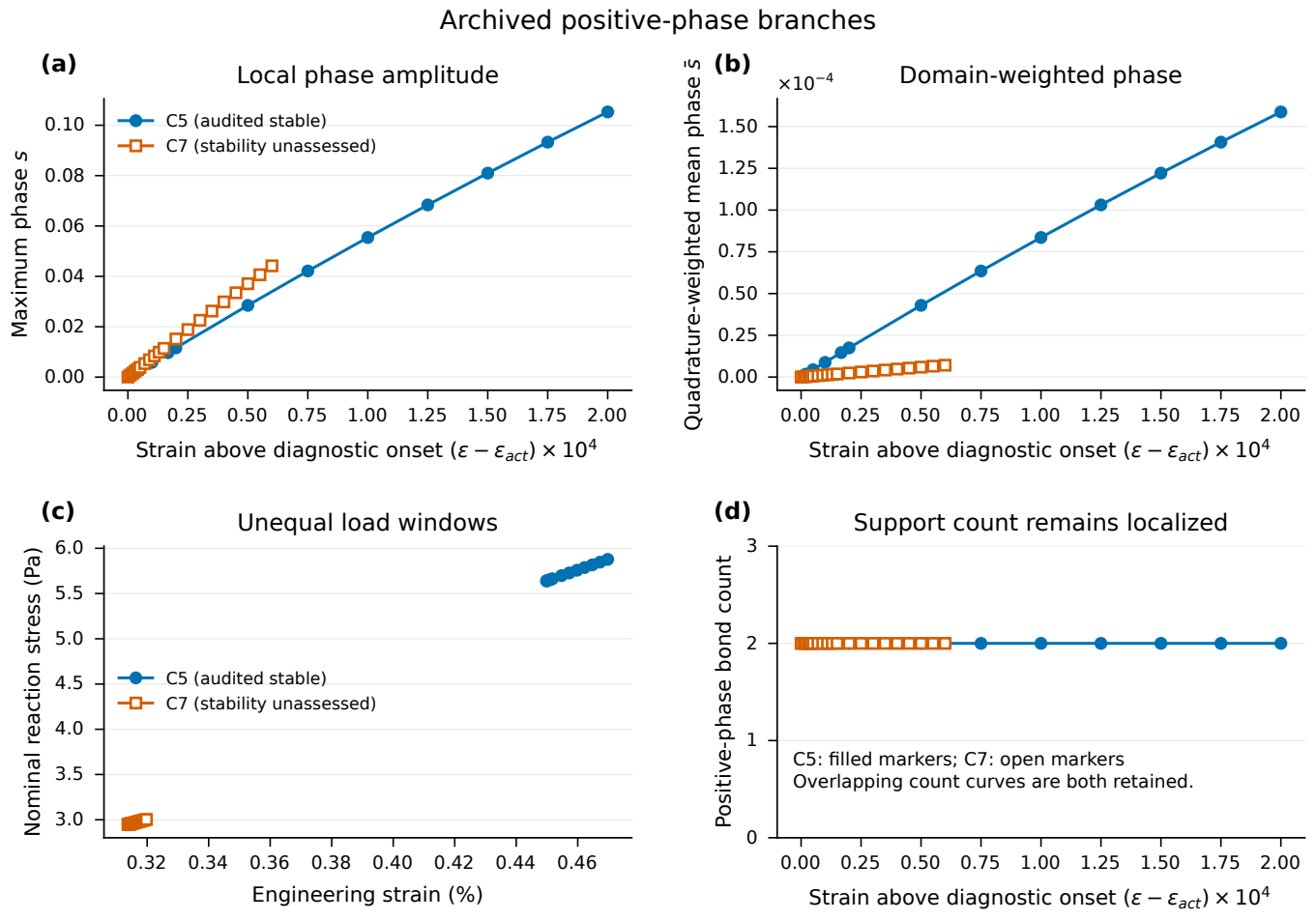


(d) Available core and declared gates

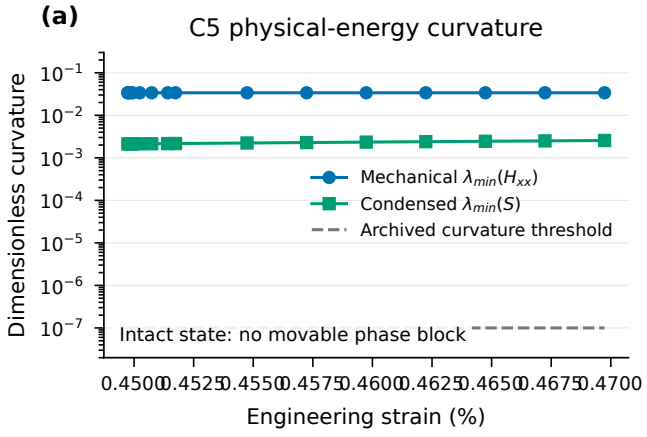
Case	1-hop family	2-hop	3-hop
C5	0	0	0
C7	0	0	0
L050	0	0	0
L075	27	0	0

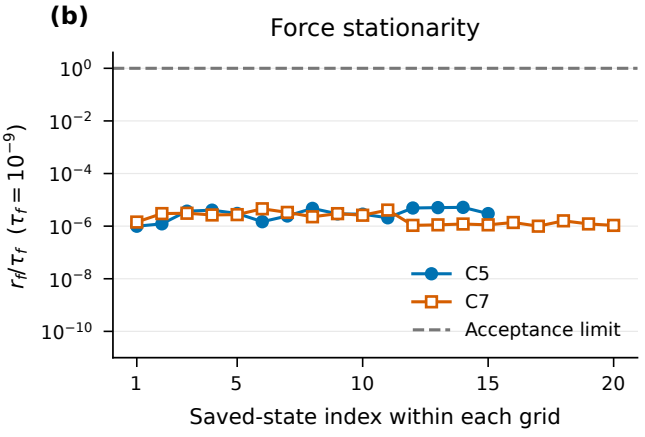
L075 full-family core: FAIL
max |f| / R = 1.07e-04 (limit 1e-10)
core / collar = 0.0200 (limit 1e-04)
Geometric completeness does not close the complete force-dependency stencil.

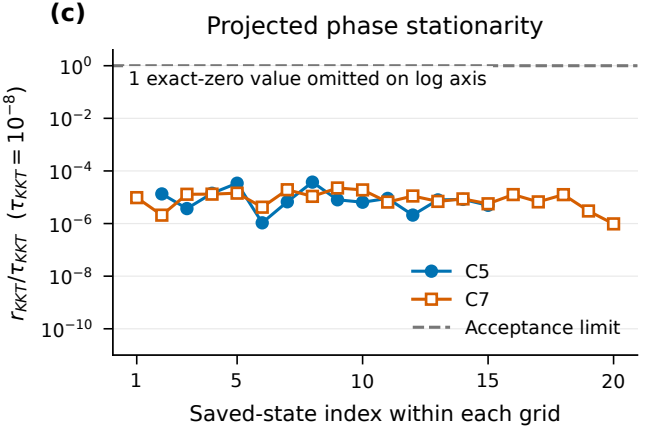

Global energy, affine virtual work and half-reaction
(a)
Affine energy error
$|U_{PD}/U_{cont} - 1|$
10^{-12}
10^{-13}
10^{-14}
10^{-15}
C5
C7
L050
L075
(b)
Virtual-work error
$|\mathcal{W}_{aff}/(R_{cont}L) - 1|$
10^{-12}
10^{-13}
10^{-14}
10^{-15}
C5
C7
L050
L075
(c)
Required half-reaction
R_{half}/R_{cont}
1% audit band
1.5
1.4
1.3
1.2
1.1
1.0
1.376
1.442
1.098
1.002
C5
C7
L050
L075


Archived positive-phase branches
(a)
Local phase amplitude
C5 (audited stable)
C7 (stability unassessed)
Maximum phase s
0.00
0.02
0.04
0.06
0.08
0.10
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00
Strain above diagnostic onset $(\varepsilon - \varepsilon_{act}) \times 10^4$
(b)
Domain-weighted phase
$\times 10^{-4}$
Quadrature-weighted mean phase $\bar{s}$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00
Strain above diagnostic onset $(\varepsilon - \varepsilon_{act}) \times 10^4$
(c)
Unequal load windows
C5 (audited stable)
C7 (stability unassessed)
Nominal reaction stress (Pa)
3.0
3.5
4.0
4.5
5.0
5.5
6.0
0.32 0.34 0.36 0.38 0.40 0.42 0.44 0.46
Engineering strain (%)
(d)
Support count remains localized
Positive-phase bond count
0
1
2
3
C5: filled markers; C7: open markers
Overlapping count curves are both retained.
0.00 0.25 0.50 0.75 1.00 1.25 1.50 1.75 2.00
Strain above diagnostic onset $(\varepsilon - \varepsilon_{act}) \times 10^4$

Stationarity and stability of the audited states







(d) What the evidence establishes

C5: 15 / 15 audited states stable
Includes 13 positive-phase states.
No instability identified in this window.

C7: 20 stationary states
Stability: NOT ASSESSED
Unclassified does not mean unstable.

No instability interval or rupture criterion has been established.

What current evidence implies for future clot simulation

Existing evidence	Interpretation limit	Development implication
Two first-set bonds on each grid; onset shifts by -30.2% (C7 vs C5).	Bond counts and sampling are not a physical crack area.	Define and validate a failure observable before rupture claims.
C5: 13 stable positive states; C7 stability remains unassessed.	Phase growth does not establish instability or propagation.	Keep stationarity, stability and physical failure tests separate.
Half-reaction ratios: C5 1.376; C7 1.442.	Global energy agreement does not validate physical traction.	Validate finite-domain loading and its work-conjugate forces.
L075: 27 full-family particles; 0 three-hop closed particles.	The geometric core test fails; root cause is inconclusive.	Use a complete force-support oracle before assigning a cause.
Current results are numerical coupon and audit evidence.	No clot-specific calibration or biological rupture is established.	Relate validated mechanics to clot data, then assess coupling.

Development implications, not completed experiments or patient-level predictions.